

# Lecture Notes on Differential Equations and Transform Techniques

*Lecture Notes for Spring 2025*

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

## References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

## EE1060: Differential Equations and Transform Techniques

### Lecture Notes - 07

#### Topics covered :

1. Implicit Methods for Numerical Solution of Differential Equations

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- Consider a first-order ODE

$$\frac{dy}{dx} = f(x, y) \quad (1)$$

It is not always guaranteed that an analytical solution to equation (1) exists. Even if one does exist, it may not be easy to compute. One useful way to analyze ODEs of the form (1) is by using [slope fields](#) (or [direction fields](#)) [1]. Direction fields provide a way to visualize the behavior of the solution to a differential equation without solving it explicitly. Moreover, direction fields can provide insights into the nature of the solution, such as its steady states, oscillatory behavior, and stability.

A direction field consists of short line segments (called tangents) drawn at evenly spaced points in the  $xy$ -plane. The slope of each line segment at a point  $(x_i, y_i)$  is given by  $f(x_i, y_i)$ . Generating a direction field for an ODE of the form (1) typically involves the following steps:

1. Choose a grid of evenly spaced points in the  $xy$ -plane.
2. At each point  $(x_i, y_i)$ , compute the slope  $m = f(x_i, y_i)$ .
3. At each point, draw a short line segment with slope  $m$ , indicating the direction of the solution at that point.

In contrast, an [isocline](#) is a curve where the slope of the solution to the differential equation is constant. For an ODE of the form (1), an isocline is defined by the equation

$$f(x, y) = c \quad (2)$$

where  $c$  is a constant. The isocline represents the level set of the function  $f(x, y)$  corresponding to the value  $c$ . The isocline method is a more efficient way of plotting direction fields, especially when done by hand.

The steps for using the isocline method to plot the direction field are:

1. For a given differential equation, determine the curves where the slope is constant. For example, if  $f(x, y) = x + y$ , the isoclines are straight lines given by  $y = c - x$ .
2. For each constant  $c$ , draw the corresponding isocline. These curves represent the locations where the slope of the solution is  $c$ .
3. For each isocline, the slope of the solution at every point on that curve is constant. Plot small line segments along the isocline to represent the direction field.

The isocline method provides a more straightforward way to visualize the behavior of the solution, especially when the direction field is complex or the solution's structure is not easily discernible.

- Consider the initial value problem (IVP):

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad (3)$$

The **Forward Euler** method is an explicit method for solving the IVP. It approximates the solution by discretizing the solution over small time steps and using the tangent slopes at the **current step**.

Given  $y_0$  at  $x_0$ , the goal is to approximate the solution at the next time step  $x_1 = x_0 + h$ , where  $h$  is the step size. The key idea behind the Forward Euler method is to use the slope of the solution at the current point  $(x_0, y_0)$  to update our approximation for the next point  $(x_1, y_1)$  (see Fig. 1). In the forward Euler method, the solution at

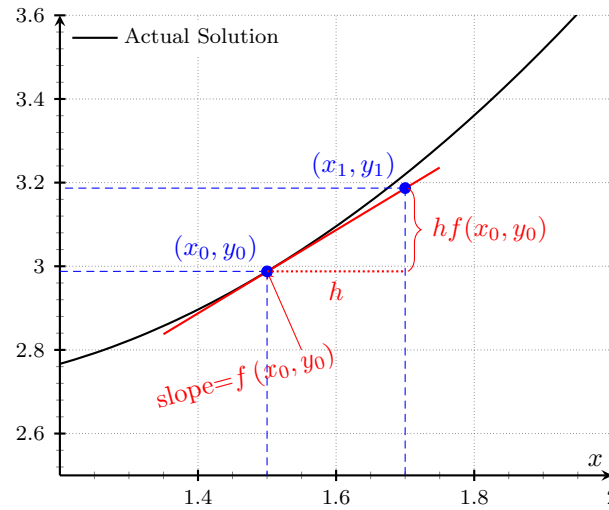


Fig. 1: Illustration of the Forward Euler method.

$x_1$  is thus approximated as

$$y_1 = y_0 + hf(x_0, y_0) \quad (4)$$

Once  $y_1$  is computed, the slope at the point  $(x_1, y_1)$ , which is  $f(x_1, y_1)$  can be computed, which can further be used as the starting point for the next time step, and the process continues. The general update equation for the Forward Euler method is:

$$y_{n+1} = y_n + hf(x_n, y_n) \quad (5)$$

In this method, the value of  $y_{n+1}$  is calculated directly using the known values at the current point. This makes the Forward Euler method an **explicit** method, where the next value depends explicitly on the current value of the function and its derivative.

- The **local error** in a numerical method refers to the error made in a single step of the method. For the Forward Euler method, the local error at each step can be derived by considering the Taylor series expansion of the exact solution. Suppose the exact solution be  $y = g(x)$ . Then according to Taylor's series, the exact solution can be computed as

$$g(x_0 + h) = g(x_0) + h \left. \frac{dg}{dx} \right|_{x_0} + \frac{h^2}{2} \left. \frac{d^2g}{dx^2} \right|_{x_0} + O(h^3) = y_0 + hf(x_0, y_0) + \frac{h^2}{2} \left. \frac{d^2g}{dx^2} \right|_{x_0} + O(h^3) \quad (6)$$

The value at  $x_1$  i.e.,  $y_1$  obtained using the Forward Euler method (approximate solution) is given by:

$$y_1 = y_0 + hf(x_0, y_0) \quad (7)$$

The local error  $\epsilon$  (i.e., difference between the exact solution and the numerical approximation) is given by

$$\epsilon = g(x_0 + h) - y_1 = \frac{h^2}{2} \frac{d^2y}{dx^2} \Big|_{x_0} + O(h^3) \quad (8)$$

Thus, the local error of the Forward Euler method is of order  $O(h^2)$ .

- Consider the initial value problem (IVP):

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad (9)$$

The **Runge-Kutta 2 (RK2)** method is an explicit method for solving the IVP. It approximates the solution by discretizing the problem over small time steps and using tangent slopes at two distinct points: one at the current step, and the other at a point obtained by advancing a certain distance along the slope at the current step (as shown in Fig. 2).

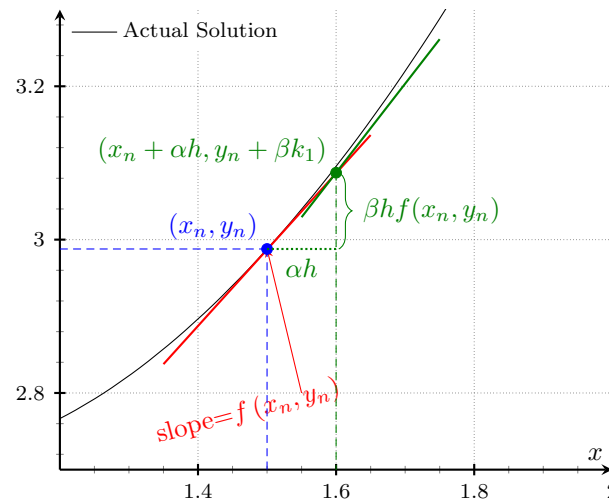


Fig. 2: Illustration of the RK2 method.

The general update equation for the RK2 method is

$$y_{n+1} = y_n + ak_1 + bk_2 \text{ where } k_1 = hf(x_n, y_n) \text{ and } k_2 = hf(x_n + \alpha h, y_n + \beta k_1) \quad (10)$$

By applying the Taylor series expansion to a function of multiple variables,  $f(x_n + \alpha h, y_n + \beta k_1)$  can be written as

$$f(x_n + \alpha h, y_n + \beta k_1) = f(x_n, y_n) + \alpha h \frac{\partial f(x_n, y_n)}{\partial x} + \beta h f(x_n, y_n) \frac{\partial f(x_n, y_n)}{\partial y} \quad (11)$$

Substituting (11) in (10) and rearranging the equation results in

$$y_{n+1} = y_n + (a+b)hf(x_n, y_n) + b\alpha h^2 \frac{\partial f(x_n, y_n)}{\partial x} + b\beta h^2 f(x_n, y_n) \frac{\partial f(x_n, y_n)}{\partial y} \quad (12)$$

The constants  $a$ ,  $b$ ,  $\alpha$  and  $\beta$  are chosen to minimize the local error. In particular, in RK2 method, the constants are derived such that the local error becomes  $O(h^3)$ . For the local error to be  $O(h^3)$ , the resulting update equation (12) must match the first three terms of the Taylor series expansion of  $g(x)$  (which is the exact solution) around  $x_n$  i.e.,

$$g(x_n + h) = g(x_n) + hg'(x_n) + \frac{h^2}{2}g''(x_n) + O(h^3) \quad (13)$$

Equation (13) can be rewritten as (under the assumption that at step  $n$  the error is zero i.e.,  $y_n = g(x_n)$ )

$$g(x_n + h) = y_n + hf(x_n, y_n) + \frac{h^2}{2}f'(x_n, y_n) + O(h^3) \quad (14)$$

Since  $y$  is a function of  $x$ ,

$$f'(x_n, y_n) = \frac{\partial f(x_n, y_n)}{\partial x} + \frac{\partial f(x_n, y_n)}{\partial y} \frac{dy}{dx} = \frac{\partial f(x_n, y_n)}{\partial x} + \frac{\partial f(x_n, y_n)}{\partial y} f(x_n, y_n) \quad (15)$$

Substituting (15) in (14) and rearranging gives

$$g(x_n + h) = y_n + hf(x_n, y_n) + \frac{h^2}{2} \frac{\partial f(x_n, y_n)}{\partial x} + \frac{h^2}{2} f(x_n, y_n) \frac{\partial f(x_n, y_n)}{\partial y} + O(h^3) \quad (16)$$

Comparing (16) and (12) gives the set of conditions the constants  $a$ ,  $b$ ,  $\alpha$  and  $\beta$  must satisfy for the local error to be  $O(h^3)$ . The conditions are

$$a + b = 1, \quad b\alpha = \frac{1}{2}, \quad \text{and } b\beta = \frac{1}{2} \quad (17)$$

There are several possible choices for the constants  $a$ ,  $b$ ,  $\alpha$  and  $\beta$  [2]. For this course, the constants are chosen to be

$$a = b = \frac{1}{2} \quad \text{and} \quad \alpha = \beta = 1 \quad (18)$$

Thus the update equation for the RK2 method is

$$y_{n+1} = y_n + \frac{1}{2}k_1 + \frac{1}{2}k_2 \quad \text{where } k_1 = hf(x_n, y_n) \quad \text{and } k_2 = hf(x_n + h, y_n + k_1) \quad (19)$$

- In the [Runge-Kutta 4 \(RK4\)](#) method, the solution at the next step is obtained using tangent slopes at four distinct points: one at the current step, two at intermediate points, and one at a point obtained by advancing a certain distance along the slope at the previous step. The idea behind RK4 is to combine these slopes in a weighted average to obtain a higher-order approximation. The update equation for the RK4 method is

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (20)$$

where

$$k_1 = hf(x_n, y_n), \quad k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \quad k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \quad k_4 = hf(x_n + h, y_n + k_3) \quad (21)$$

The terms  $k_1, k_2, k_3, k_4$  represent the slopes at the current step and at three intermediate points along the curve. More precisely,

- $k_1$  is the slope at the current point  $(x_n, y_n)$ ,
- $k_2$  and  $k_3$  are estimates of the slope at intermediate points based on the initial slope  $k_1$
- $k_4$  is the slope at the point  $(x_n + h, y_n + k_3)$ , which is the final step after advancing along the curve.

The local error in the Runge-Kutta 4th order (RK4) method is of order  $O(h^5)$ .

## References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.
- [2] R. Kaw, “Numerical methods with applications.” [Online]. Available: [https://math.libretexts.org/Workbench/Numerical\\_Methods\\_with\\_Applications\\_\(Kaw\)](https://math.libretexts.org/Workbench/Numerical_Methods_with_Applications_(Kaw))